CSC370 PaperTrade, Lossless Join Proof

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Since our database is already in BCNF we need not prove a lossless join, but for the sake of proving course level competencies, we will proceed with a hypothetical flat relation as follows:

$$\text{UserHoldings(user_id, name, email, ticker, quantity, average_price)}$$

With the complete set of attributes being:

$$C = \{\text{user_id, name, email, ticker, quantity, average_price}\}$$

We can now assert the following functional dependencies, (F):

- FD_1 : user_id $\rightarrow$ name, email
- FD_2 : user_id, ticker $\rightarrow$ quantity, average_price

We can now derive the following candidate key by computing the attribute closures:

1. **Closure of** {user_id}:
$$\{\text{user_id}\}^+ = \{\text{user_id, name, email}\} \neq C$$
Since the closure does not contain all attributes of C, {user_id} is not a superkey.
2. **Closure of** {ticker}:
$$\{\text{ticker}\}^+ = \{\text{ticker}\} \neq C$$
Since the closure does not contain all attributes of C, {ticker} is not a superkey.
3. **Closure of** {user_id, ticker}:

$$\begin{aligned} \{\text{user_id, ticker}\}^+ &\supseteq \{\text{user_id, ticker}\} && \text{(by triviality)} \\ &\supseteq \{\text{user_id, ticker, name, email}\} && \text{(applying } FD_1\text{)} \\ &\supseteq \{\text{user_id, ticker, name, email, quantity, average_price}\} && \text{(applying } FD_2\text{)} \\ &= C \end{aligned}$$

Because {user_id, ticker}$^+ = C$ and no proper subset of {user_id, ticker} determines all attributes in C, {user_id, ticker} is the minimal candidate key for UserHoldings.

BCNF Violation Analysis

A relation R with attribute set C is in **Boyce-Codd Normal Form (BCNF)** if and only if for every non-trivial functional dependency $X \rightarrow Y$, the antecedent X is a superkey of R ($X^+ = C$).

Evaluating FD_1 : user_id $\rightarrow$ name, email:

- **Antecedent (X):** {user_id}
- **Closure (X^+):** {user_id}$^+$ = {user_id, name, email} $\neq C$

Since {user_id} is not a superkey for UserHoldings, FD_1 violates BCNF, causing data redundancy as well as update, insertion, and deletion anomalies.

BCNF Decomposition Algorithm

We apply the formal BCNF decomposition algorithm (BCNF_Decomp) to eliminate the violation:

1. Select the BCNF violation FD_1 : user_id $\rightarrow$ name, email with antecedent X = {user_id}.
2. Compute the closure X^+ =
    1. **Closure of** {user_id}:
    $$\{\text{user_id}\}^+ = \{\text{user_id, name, email}\} \neq C$$
    Since {user_id}$^+ \neq C$, {user_id} is not a superkey.
    2. **Closure of** {ticker}:
    $$\{\text{ticker}\}^+ = \{\text{ticker}\} \neq C$$
    Since {ticker}$^+ \neq C$, {ticker} is not a superkey.
    3. **Closure of Composite Set** {user_id, ticker}:

$$\begin{aligned} \{\text{user_id, ticker}\}^+ &\supseteq \{\text{user_id, ticker}\} && \text{(by triviality)} \\ &\supseteq \{\text{user_id, ticker, name, email}\} && \text{(applying } FD_1\text{)} \\ &\supseteq \{\text{user_id, ticker, name, email, quantity, average_price}\} && \text{(applying } FD_2\text{)} \\ &= C \end{aligned}$$

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BCNF Violation Analysis

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Evaluating FD_1 : user_id $\rightarrow$ name, email:

- **Antecedent (X):** {user_id}
- **Closure (X^+):** {user_id}$^+$ = {user_id, name, email} $\neq C$

Since {user_id} is not a superkey for UserHoldings, FD_1 represents a **BCNF violation**, introducing data redundancy as well as update, insertion, and deletion anomalies.

BCNF Decomposition Algorithm

We apply the formal BCNF decomposition algorithm (BCNF_Decomp) to eliminate the violation:

1. Select the BCNF violation FD_1 : user_id $\rightarrow$ name, email with antecedent $X = \{\text{user_id}\}$.
2. Compute the closure $X^+ = \{\text{user_id}\}^+ = \{\text{user_id, name, email}\}$.
3. Form relation $R_1 := X^+$:

$$R_1 = \{\text{user_id, name, email}\} \implies \text{User(user_id, name, email)}$$

4. Form relation $R_2 := (C \setminus X^+) \cup X$:

$$R_2 = \{\text{user_id, ticker, quantity, average_price}\} \implies \text{Holds(user_id, ticker, quantity, average_price)}$$

Both sub-relations R_1 and R_2 now satisfy BCNF because their respective determinants are superkeys within their relation schemas.

Lossless Join Proof

According to the **Lossless Join Theorem**, a decomposition of relation R into sub-relations R_1 and R_2 is **lossless** if and only if the shared attributes $(R_1 \cap R_2)$ form a superkey for R_1 or R_2:

$$(R_1 \cap R_2) \rightarrow R_1 \quad \text{or} \quad (R_1 \cap R_2) \rightarrow R_2 \in F^+$$

1. **Attribute Intersection:**

$$R_1 \cap R_2 = \{\text{user_id, name, email}\} \cap \{\text{user_id, ticker, quantity, average_price}\} = \{\text{user_id}\}$$

2. **Closure of Shared Attributes:**

$$\{\text{user_id}\}^+ = \{\text{user_id, name, email}\} = R_1$$

3. **Superkey Verification:** Since $\{\text{user_id}\}^+ = R_1$, the shared attribute set $\{\text{user_id}\}$ forms a superkey for R_1 (User).

Conclusion: Because $(R_1 \cap R_2) \rightarrow R_1$, the decomposition of UserHoldings into R_1 (User) and R_2 (Holds) satisfies the Lossless Join Theorem. Rejoining the relations via natural join is mathematically proven to be lossless:

$$\pi_{R_1}(\text{UserHoldings}) \bowtie \pi_{R_2}(\text{UserHoldings}) = \text{UserHoldings}$$